

Fundamentals of Data Science Lab Programs Set (1 – 6)

1. Average score for each subject and highest average

Python code :

```
import numpy as np

student_scores = np.array([
    [80, 75, 90, 85],
    [70, 80, 85, 75],
    [90, 85, 80, 90],
    [85, 90, 95, 80]
])

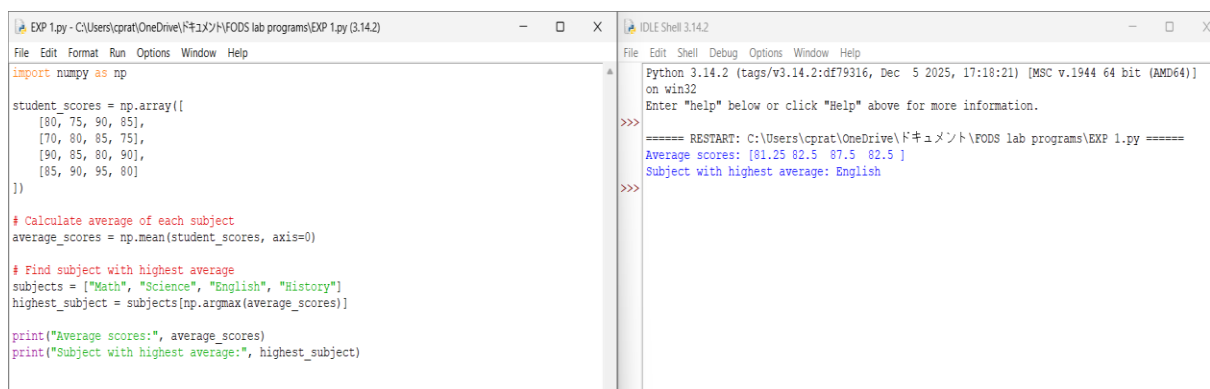
# Calculate average of each subject
average_scores = np.mean(student_scores, axis=0)

# Find subject with highest average
subjects = ["Math", "Science", "English", "History"]
highest_subject = subjects[np.argmax(average_scores)]

print("Average scores:", average_scores)

print("Subject with highest average:", highest_subject)
```

Output :

The screenshot shows a Python IDE with two windows. The left window, titled 'EXP 1.py', contains the Python code for calculating average scores and finding the subject with the highest average. The right window, titled 'IDLE Shell 3.14.2', shows the output of the code. The output consists of two lines: 'Average scores: [81.25 82.5 87.5 82.5]' and 'Subject with highest average: English'.

```
File Edit Format Run Options Window Help
import numpy as np

student_scores = np.array([
    [80, 75, 90, 85],
    [70, 80, 85, 75],
    [90, 85, 80, 90],
    [85, 90, 95, 80]
])

# Calculate average of each subject
average_scores = np.mean(student_scores, axis=0)

# Find subject with highest average
subjects = ["Math", "Science", "English", "History"]
highest_subject = subjects[np.argmax(average_scores)]

print("Average scores:", average_scores)

print("Subject with highest average:", highest_subject)
```

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\cprat\OneDrive\ドキュメント\FODS lab programs\EXP 1.py =====
Average scores: [81.25 82.5 87.5 82.5 ]
Subject with highest average: English
>>>
```

2. Average price of all products sold

Python code :

```
import numpy as np
```

```

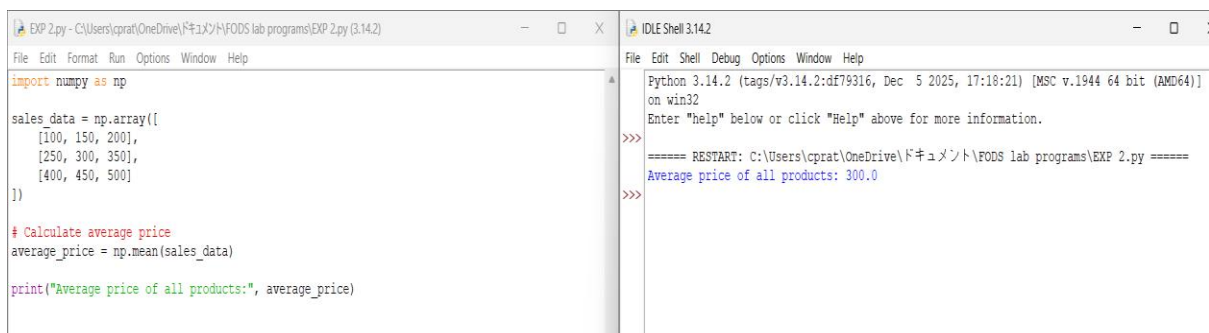
sales_data = np.array([
    [100, 150, 200],
    [250, 300, 350],
    [400, 450, 500]
])

# Calculate average price
average_price = np.mean(sales_data)

print("Average price of all products:", average_price)

```

Output :



The screenshot shows a Python IDE with two windows. The left window, titled 'EXP 2.py', contains the following code:

```

import numpy as np

sales_data = np.array([
    [100, 150, 200],
    [250, 300, 350],
    [400, 450, 500]
])

# Calculate average price
average_price = np.mean(sales_data)

print("Average price of all products:", average_price)

```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the code:

```

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\cprat\OneDrive\ドキュメント\FODS lab programs\EXP 2.py =====
Average price of all products: 300.0
>>>

```

3. Average sale price of houses with more than 4 bedrooms

Python code :

```

import numpy as np

house_data = np.array([
    [3, 1200, 200000],
    [5, 1800, 350000],
    [4, 1500, 280000],
    [6, 2200, 450000],
    [5, 2000, 400000]
])

# Select houses with more than 4 bedrooms
selected_houses = house_data[house_data[:, 0] > 4]

# Get sale prices
sale_prices = selected_houses[:, 2]

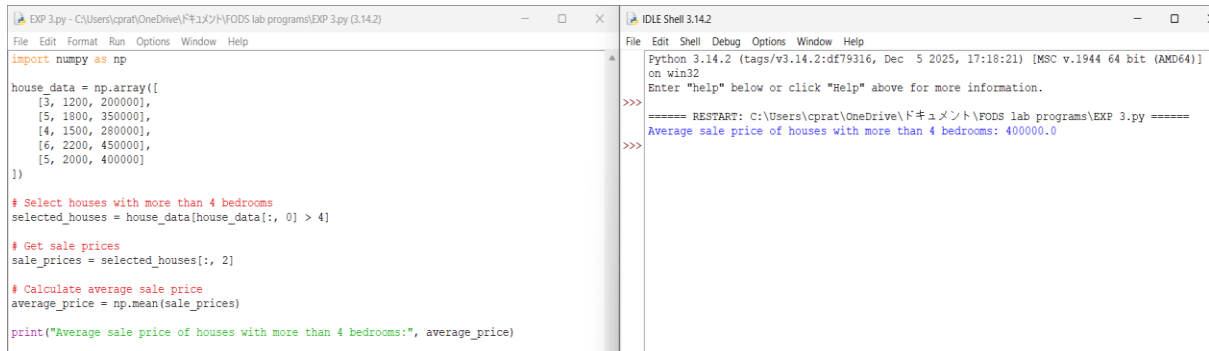
```

```
# Calculate average sale price
```

```
average_price = np.mean(sale_prices)
```

```
print("Average sale price of houses with more than 4 bedrooms:", average_price)
```

Output :



The screenshot shows a Python IDE with two windows. The left window displays the following code:

```
import numpy as np

house_data = np.array([
    [3, 1200, 200000],
    [5, 1800, 350000],
    [4, 1500, 280000],
    [6, 2200, 450000],
    [5, 2000, 400000]
])

# Select houses with more than 4 bedrooms
selected_houses = house_data[house_data[:, 0] > 4]

# Get sale prices
sale_prices = selected_houses[:, 2]

# Calculate average sale price
average_price = np.mean(sale_prices)

print("Average sale price of houses with more than 4 bedrooms:", average_price)
```

The right window shows the output:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:\Users\cprat\OneDrive\ドキュメント\FODS lab programs\EXP 3.py =====
Average sale price of houses with more than 4 bedrooms: 400000.0
>>>
```

4. Total yearly sales and percentage increase

Python code :

```
import numpy as np
```

```
sales_data = np.array([100000, 120000, 130000, 150000])
```

```
# Calculate total sales
```

```
total_sales = np.sum(sales_data)
```

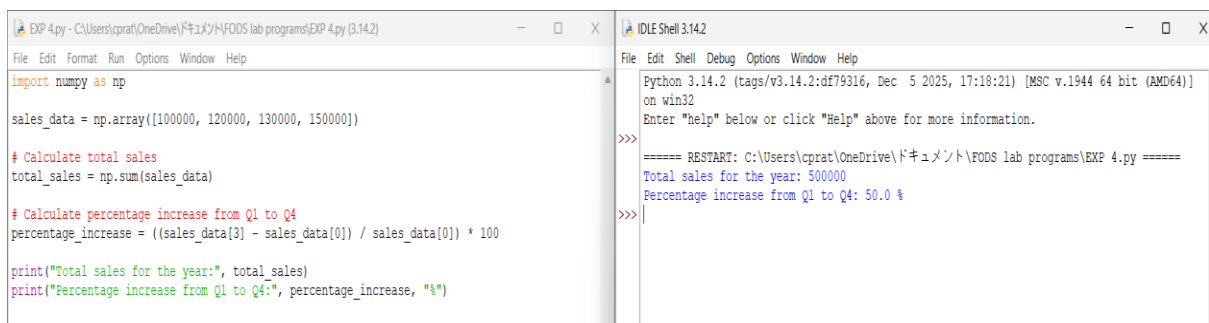
```
# Calculate percentage increase from Q1 to Q4
```

```
percentage_increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100
```

```
print("Total sales for the year:", total_sales)
```

```
print("Percentage increase from Q1 to Q4:", percentage_increase, "%")
```

Output :



The screenshot shows a Python IDE with two windows. The left window displays the following code:

```
import numpy as np

sales_data = np.array([100000, 120000, 130000, 150000])

# Calculate total sales
total_sales = np.sum(sales_data)

# Calculate percentage increase from Q1 to Q4
percentage_increase = ((sales_data[3] - sales_data[0]) / sales_data[0]) * 100

print("Total sales for the year:", total_sales)
print("Percentage increase from Q1 to Q4:", percentage_increase, "%")
```

The right window shows the output:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:\Users\cprat\OneDrive\ドキュメント\FODS lab programs\EXP 4.py =====
Total sales for the year: 500000
Percentage increase from Q1 to Q4: 50.0 %
>>>
```

5. Average fuel efficiency and percentage improvement

Python code :

```
import numpy as np
```

```

fuel_efficiency = np.array([25, 30, 35, 40, 45])

# Calculate average fuel efficiency
average_efficiency = np.mean(fuel_efficiency)

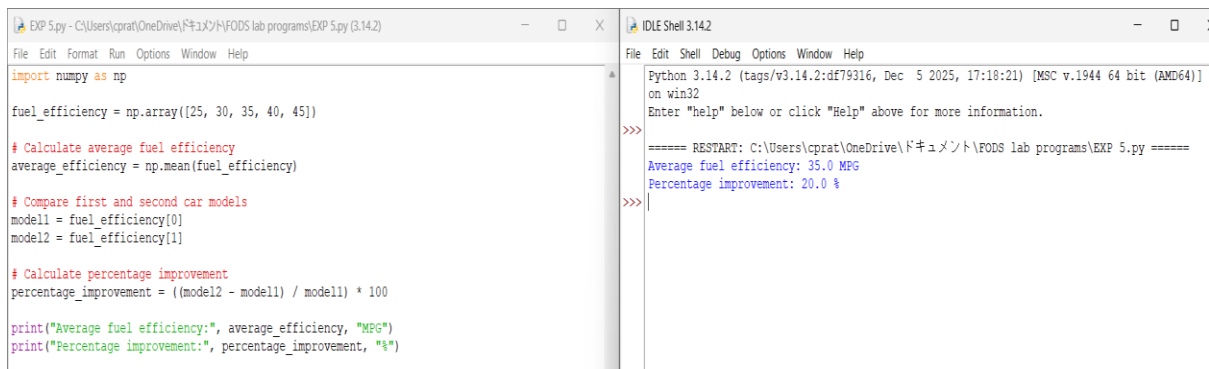
# Compare first and second car models
model1 = fuel_efficiency[0]
model2 = fuel_efficiency[1]

# Calculate percentage improvement
percentage_improvement = ((model2 - model1) / model1) * 100

print("Average fuel efficiency:", average_efficiency, "MPG")
print("Percentage improvement:", percentage_improvement, "%")

```

output :



The screenshot shows a Python IDE with two windows. The left window displays the Python code from the previous block. The right window shows the output of the code, which includes the average fuel efficiency and the percentage improvement.

```

EXP 5.py - C:\Users\cprat\OneDrive\ドキュメント\FODS lab programs\EXP 5.py (3.14.2)
File Edit Format Run Options Window Help
import numpy as np
fuel_efficiency = np.array([25, 30, 35, 40, 45])
# Calculate average fuel efficiency
average_efficiency = np.mean(fuel_efficiency)
# Compare first and second car models
model1 = fuel_efficiency[0]
model2 = fuel_efficiency[1]
# Calculate percentage improvement
percentage_improvement = ((model2 - model1) / model1) * 100
print("Average fuel efficiency:", average_efficiency, "MPG")
print("Percentage improvement:", percentage_improvement, "%")

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\cprat\OneDrive\ドキュメント\FODS lab programs\EXP 5.py =====
Average fuel efficiency: 35.0 MPG
Percentage improvement: 20.0 %
>>>

```

6. Total cost including discount and tax

Python code :

```

prices = [100, 200, 300]
quantities = [2, 1, 3]
discount_rate = 10
tax_rate = 5

# Calculate total before discount
subtotal = sum(p * q for p, q in zip(prices, quantities))

# Calculate discount
discount = subtotal * discount_rate / 100

# Calculate amount after discount
amount_after_discount = subtotal - discount

# Calculate tax

```

```
tax = amount_after_discount * tax_rate / 100
```

```
# Calculate final cost
```

```
total_cost = amount_after_discount + tax
```

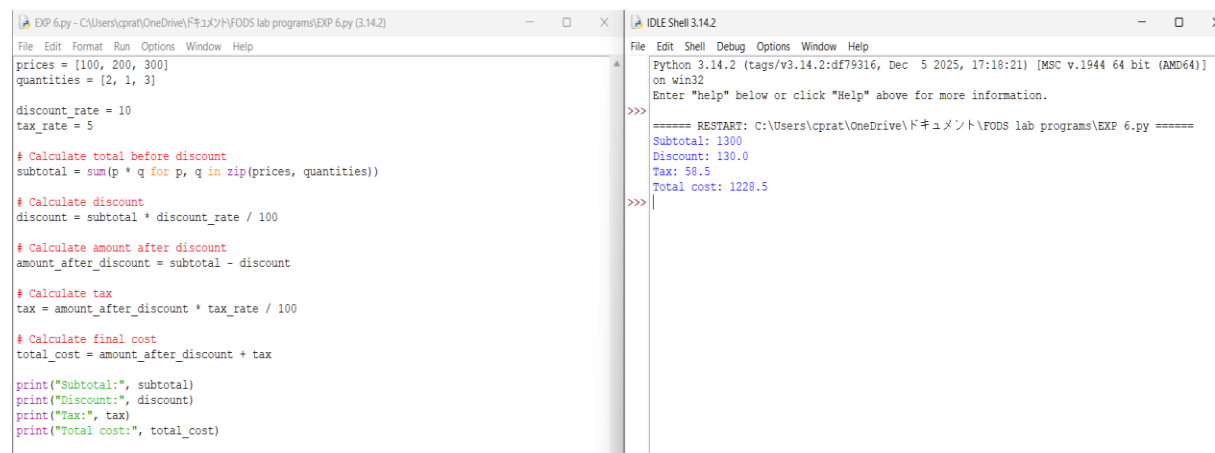
```
print("Subtotal:", subtotal)
```

```
print("Discount:", discount)
```

```
print("Tax:", tax)
```

```
print("Total cost:", total_cost)
```

Output :



The screenshot shows a Python IDE with two windows. The left window, titled 'EXP 6.py', contains the following code:

```
prices = [100, 200, 300]
quantities = [2, 1, 3]

discount_rate = 10
tax_rate = 5

# Calculate total before discount
subtotal = sum(p * q for p, q in zip(prices, quantities))

# Calculate discount
discount = subtotal * discount_rate / 100

# Calculate amount after discount
amount_after_discount = subtotal - discount

# Calculate tax
tax = amount_after_discount * tax_rate / 100

# Calculate final cost
total_cost = amount_after_discount + tax

print("Subtotal:", subtotal)
print("Discount:", discount)
print("Tax:", tax)
print("Total cost:", total_cost)
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the program:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\cprat\OneDrive\ドキュメント\FODS lab programs\EXP 6.py =====
Subtotal: 1300
Discount: 130.0
Tax: 59.5
Total cost: 1228.5
>>>
```